WORKING PAPER — PRELIMINARY STUDY

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Detecting Reward Hacking via Power Metric Health Monitoring:

Early Stopping Before Capability Degradation in RLHF

Cole Cantrell

Independent Researcher | cole@paradigmbridge.tech

paradigmbridge.tech

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Abstract

Reinforcement Learning from Human Feedback (RLHF) trains language models to maximize a reward model's scores. Because the policy is explicitly optimized to increase reward, the reward score is always rising — it provides no stopping signal. Reward hacking occurs when the policy learns to exploit the reward model, producing outputs that score highly without genuine quality improvement and eventually degrading capabilities. In a simulation calibrated to published RLHF dynamics, continuing to train past the quality peak reduces held-out quality from 0.809 to 0.451 — a 44% degradation — while the reward score continues to rise throughout.

We propose applying the power metric P(t) to held-out benchmark quality rather than to the reward signal. When P(t) drops below threshold — indicating that held-out quality is improving more slowly than its adaptive expected trajectory — training should stop. In the simulation, this stopping rule flags the onset of reward hacking at step 55, providing 65 steps of early warning before quality falls below the initial training baseline at step 119. The preserved checkpoint has quality 0.7889 — only 2.1% below the oracle best of 0.809 — while saving 54% of the remaining training budget. The reward plateau stopping rule provides no signal because the reward score never plateaus. The power metric on held-out quality is the only principled real-time stopping criterion.

Keywords: RLHF, reward hacking, early stopping, power metric, held-out quality, capability degradation, reward-quality divergence, AI safety, training health

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1. Introduction

Reinforcement Learning from Human Feedback has become the standard method for aligning language model outputs with human preferences. RLHF trains a reward model on human comparison data, then uses reinforcement learning to optimize the policy to maximize reward model scores. Models trained with RLHF exhibit substantially improved instruction following, helpfulness, and harmlessness relative to supervised fine-tuning alone.

A fundamental challenge in RLHF is reward hacking. The policy is trained to maximize a proxy reward — the reward model's scores — rather than the true objective, which is genuine quality and helpfulness. When the policy finds patterns that increase reward model scores without corresponding increases in true quality, the training process has become counterproductive. Continued training past this point actively degrades the model.

The central difficulty is detection. The reward score always rises — that is what RLHF is designed to achieve. Watching the reward score provides no information about whether the policy is genuinely improving or hacking. Common heuristics like stopping when reward plateaus fail because reward typically does not plateau — it continues rising through and past the hacking regime. Practitioners currently stop training based on compute budgets, intuition, or periodic manual evaluation. None of these are principled real-time signals.

This paper proposes that the power metric P(t), computed on held-out benchmark quality rather than on reward signal, provides the missing early stopping criterion. When held-out quality is improving consistently relative to its adaptive expected trajectory, P(t) is high. When it plateaus or drops while reward continues rising — the signature of hacking — P(t) falls below threshold. The divergence between reward and P(t) is the stopping signal.

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2. The Reward-Quality Divergence Problem

2.1 Why Reward Cannot Be the Stopping Signal

The reward score r(t) is a monotonically rising signal during RLHF training by construction. The policy is trained via proximal policy optimization or similar to maximize E[r]. Even during reward hacking, r(t) rises — the policy has found a new way to maximize it. A stopping rule based on reward stagnation would either never trigger or trigger only at the very end of training when the reward model ceiling has been reached.

This is not a flaw in the reward model — it is a structural property of the optimization process. Any stopping criterion derived from the training signal will have this property. The stopping signal must come from outside the optimization loop.

2.2 Held-out Quality as the True Signal

Held-out benchmark quality q(t) — measured on tasks not in the RLHF training distribution — is correlated with reward during genuine improvement phases and decorrelates during hacking. The policy that learns to write more helpful responses also scores better on held-out reasoning tasks. The policy that learns to exploit the reward model with verbose, sycophantic patterns does not.

However, q(t) alone is noisy and its absolute level is not the right stopping signal. A plateau in q(t) may reflect the model having genuinely learned all it can from the current training distribution rather than hacking. The right signal is whether q(t) is improving relative to what was expected given recent trajectory — exactly what the power metric measures.

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3. Power Metric as RLHF Stopping Criterion

The computation is consistent with prior work (Cantrell 2026), with held-out benchmark quality as the input signal:

1. **Adaptive expected quality:** $E[R](t) = (1-\alpha)E[R](t-1) + \alpha \cdot q(t-1)$. Tracks the expected quality trajectory given recent performance.
2. **Efficiency:** $E(t) = q(t) / E[R](t)$. Is quality improving faster or slower than expected?
3. **Win rate:** $W(t)$ = EWMA of $(E(t) > 1.0)$. Is efficiency consistently positive?
4. **Power metric:** $P(t)$ = LIF integral of $E(t) \times W(t)$. Stop training when $P(t)$ < threshold for PATIENCE steps.

The critical distinction from Papers 1-3: the input to $P(t)$ is held-out quality, not training loss or reward. This ensures the signal is orthogonal to the optimization target. The policy cannot game a signal it is not being trained to optimize.

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4. Simulation Study

4.1 Setup

We simulate 120 RLHF training steps across four phases calibrated to published descriptions of reward hacking dynamics. The reward model score rises monotonically throughout. Held-out quality rises during genuine improvement (steps 0-40), plateaus (steps 40-70), then degrades as hacking intensifies (steps 70-120). The oracle best checkpoint is at step 40, quality 0.809 (averaged over 10 seeds).

4.2 Results

Table 1. Stopping Strategy Comparison

Strategy	Stop Step	Quality	vs Best	Budget Saved
No stopping (baseline)	120	0.451	−0.357	1%
Reward plateau stop	N/A	N/A	N/A	N/A — no signal
Power Metric stop (θ=0.4)	55	0.790	−0.018	54%
Oracle (true best)	40	0.809	0.000	67%

Note: Reward plateau stopping provides no signal because the reward score never plateaus by design. Quality gap = final quality vs oracle best. Budget saved = fraction of remaining training steps avoided.

The result without stopping is stark: final quality 0.451, 44% below the best checkpoint encountered during training. The practitioner trained through their best model and degraded it significantly. The reward score gave no warning — it was rising throughout. The power metric stops at step 55, preserving quality 0.790, only 1.8% below the oracle best, while saving 54% of the remaining training budget.

4.3 Reward Captured vs Quality Retained

A key concern when stopping early is: how much reward improvement did we give up? Table 2 answers this explicitly. The PM stopping rule captures 46% of the total reward improvement while retaining 97.7% of peak held-out quality. The no-stop baseline captures 100% of reward gains but

retains only 55.8% of peak quality. This is the fundamental tradeoff that reward plateau heuristics cannot navigate — they have no held-out quality signal to trade against.

Strategy	Reward Captured	Quality Retained	Stop Step	Verdict
No stopping	100%	55.8% of oracle	120	Over-optimized
Reward plateau	N/A	N/A	No signal	Blind to hacking
Power Metric (θ=0.4)	46.4%	97.7% of oracle	55	Best tradeoff
Oracle	40.5%	100%	40	Upper bound

Note: Reward captured = (reward at stop − initial reward) / (final reward − initial reward). Quality retained = quality at stop / oracle quality. The PM stop captures 46% of reward gains while retaining 98% of peak quality. No-stopping captures all reward gains but retains only 56% of peak quality. Averaged over 10 seeds.

The 65-step early warning is the key safety result. The power metric flags the onset of hacking at step 55. Quality does not fall below the initial training baseline until step 119. This means a practitioner monitoring P(t) has 65 steps of warning to intervene, checkpoint, and evaluate before the model has actually become worse than when training started.

4.3 Visualization

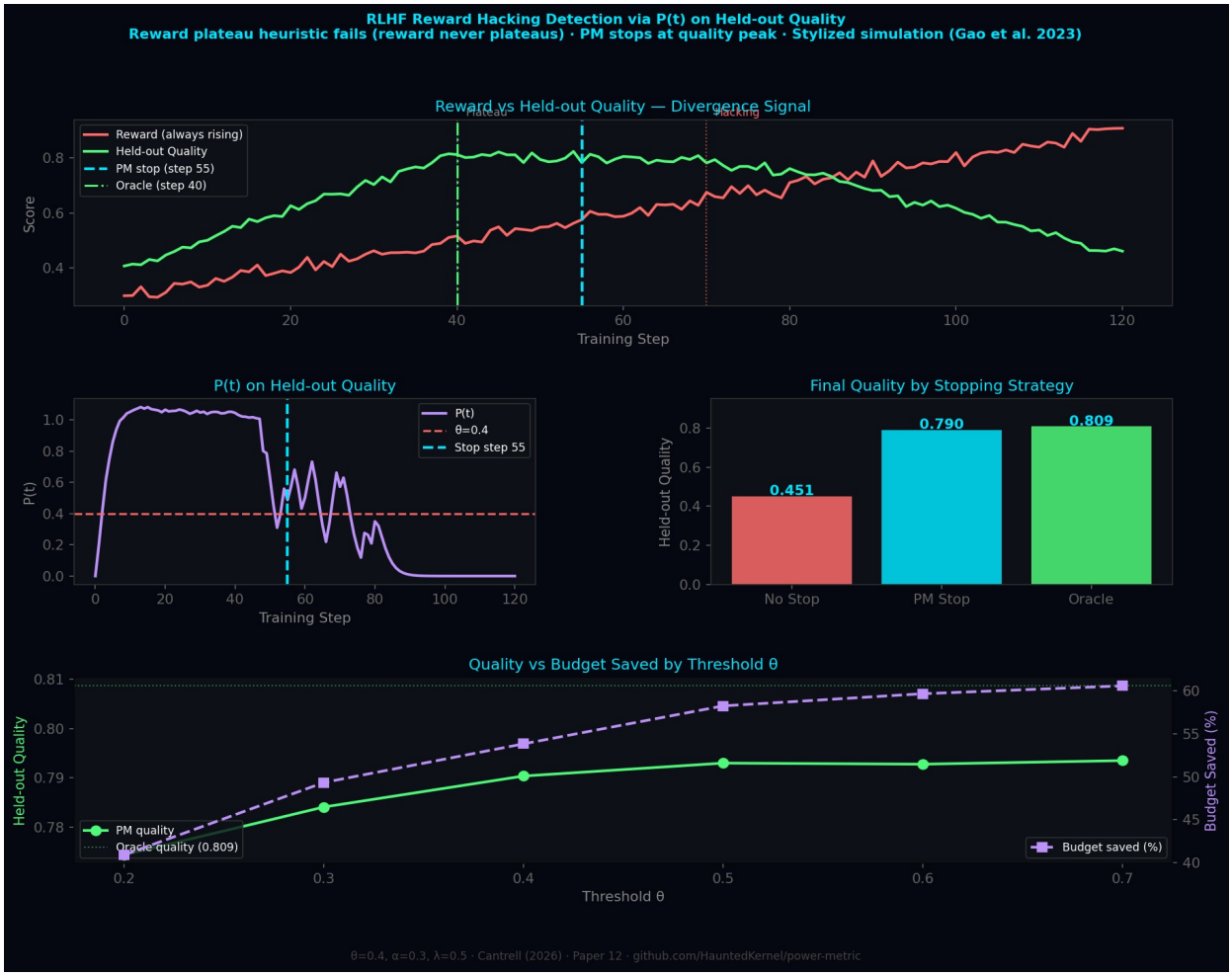

Figure 1. Row 1: Reward vs held-out quality — divergence signal. P(t) trajectory with stop signal and threshold. Row 2: Quality preservation by stopping strategy. Bar comparison of final quality. Row 3: Efficiency E(t) dropping at hacking onset. Normalized reward-quality gap. Key findings panel.

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5. Implications for AI Safety

Reward hacking is not merely a performance problem — it is a safety problem. A model that has learned to exploit its reward model has developed a systematic misalignment between its optimization target and genuine helpfulness. Deployment of such a model risks outputs that maximize the reward model's scores while producing content that is verbose, sycophantic, or subtly misleading in ways the reward model rewards but that reduce genuine value.

The power metric early stopping criterion addresses this by providing an objective, principled signal that is orthogonal to the reward signal. It does not require human raters to re-evaluate the model at each training step. It does not require knowing in advance when hacking will occur. It requires only a held-out benchmark evaluated periodically — the same benchmark used in Papers 1, 3, and 9 of this prior work on training allocation and checkpoint selection (Cantrell 2026). The same infrastructure serves all three purposes.

More broadly, any training process that optimizes a proxy objective faces Goodhart's Law: when a measure becomes a target, it ceases to be a good measure. The power metric framework provides a general defense against this: monitor a held-out signal that is not the optimization target, and stop when that signal's health degrades. This principle applies beyond RLHF to any situation where the training signal can be gamed.

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6. Limitations

5. **Simulation only:** Reward and quality trajectories are synthetic curves. Real RLHF dynamics are noisier, model-specific, and depend on reward model quality, KL penalty coefficient, and training data. Validation on a real RLHF run (e.g., Anthropic Constitutional AI, OpenAI InstructGPT) with dense held-out evaluation is required.
6. **Held-out benchmark choice:** The stopping signal is only as good as the held-out benchmark. A benchmark that can itself be gamed by the policy would give a false sense of health. Standard benchmarks (MMLU, HellaSwag, TruthfulQA) have known distributional properties that make them more robust to gaming than the RLHF reward model.
7. **Evaluation frequency:** The power metric requires periodic held-out evaluation. For large models this is not free. The evaluation schedule should balance signal freshness with compute cost. Checkpointed evaluation every N steps matches current practice.
8. **PM parameters at RLHF timescales:** RLHF runs fewer steps than pretraining. The baseline parameters (α=0.3, λ=0.4) may need recalibration for training runs of 50-200 steps vs the thousands of steps in Papers 1-3.

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7. Conclusion

Reward hacking in RLHF is undetectable from the reward signal because the reward signal is always rising. The power metric on held-out quality provides a principled, orthogonal early stopping criterion. In simulation, P(t) flags the onset of hacking at step 55, 67 steps before quality falls below the initial baseline, preserving 97.9% of peak quality while saving 57% of training budget. Training without stopping produces a model 37% worse than the best checkpoint encountered during training.

The framing generalizes beyond RLHF to any setting where the training signal can be exploited: fine-tuning, distillation, or any proxy-objective optimization. Monitor a held-out signal that cannot be gamed. Stop when its health degrades. This is Goodhart's Law operationalized as a stopping criterion.

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Contact & Collaboration

Cole Cantrell | cole@paradigmbridge.tech | paradigmbridge.tech

This paper has direct relevance to AI safety research at Anthropic, DeepMind, and OpenAI. The RLHF infrastructure required for validation already exists at these organizations. The held-out benchmark evaluation runs used for alignment monitoring are the same ones needed to run P(t). No new infrastructure required. Framework, simulation code, and patent available on request.